Field Guide: Cancer Lineage and Subtype Classification from CRISPR Dependency Profiles

MPHY 6120 — Final Project
Authors: MJ, Harry, Riza
Status: Investigational — not approved for routine clinical use
Version: 1.0 | April 2026

1. Executive Summary

The tool outlined in this guide is a Multi-Layer Perceptron which is a type of deep-learning neural network. The purpose of this tool is to classify cancer lineage and subtype from CRISPR Dependency Scores sourced from the public anonymous cancer database DepMap. These scores show how dependent a cancer sample is on a variety of different genes (500 genes were chosen among the genome for the metrics of this tool) for its proliferation and survival. The MLP learns linear and non-linear patterns between these score features themselves and the target lineage and subtype classes to eventually predict lineage and subtype based on an input of new dependency scores.

The intended users of this tool are clinical oncologists and pathologists working with real cancer patients. These users would have access to or the means to acquire such data required for this model and be in a position to find utility within the MLP to more easily and efficiently classify their patients' cancers' lineages and subtypes. It is important to note that this should be a support tool and not a standalone diagnostic tool without more advanced testing. This tool may also provide a few of the most probable lineages and subtypes which narrows the amount of biological testing the users have to do to confirm the predictions.

This MLP should be used after the CRISPR Dependency Scores have been obtained from patients and before biological testing has been done to limit the tests needed to be done to confirm predictions by the tool.

Primary performance highlights (averaged over 10 independent runs):

Task	Accuracy	Top-3 Accuracy	ROC-AUC
MLP (lineage)	~54%	~73%	~0.91
MLP (subtype)	~45–47%	~63–64%	~0.90
Most frequent baseline (lineage)	~10%	~14%	~0.50
Most frequent baseline (subtype)	~6%	~7%	~0.50

The model substantially outperforms the baseline which uses the most frequent strategy and performs exceptionally well when accounting for its' top three scores. Among other metrics (see Appendix) it has a strong ROC-AUC score (~0.90) which suggests that it has meaningful discriminative ability.

2. Clinical Context

Problem Statement

Cancer classification is an issue of high consequence in oncology which can provide high benefit to the patient. Determining both the tissue/organ of origin (lineage) and a more specific classification based on molecular patterns (subtype within a broader lineage) can heavily impact the trajectory of a patient. Such classifications can affect decisions of prognosis, treatment plan, and therapeutic vulnerabilities. However, modern methods of classification while highly effective have downfalls with the testing being time intensive, expensive, and vast. These tests include histopathology, immunohistochemistry, and genomic profiling. Commonly cited limitations on these tests include:

- **Cancers of unknown primary (CUP):** In 3–5% of metastatic cases, the tissue or organ for the cancer's origin cannot be determined with usual testing.
- **Ambiguous histology:** Tumors may have ambiguous features requiring extensive testing and specialist's inputs.
- **Functional information gap:** Standard diagnostics reveal molecular information such as proteins present, but little information on genomic dependencies of the cancer which provide crucial insight for drug vulnerabilities.

What CRISPR Dependency Profiling Offers

CRISPR Dependency Scores are determined per gene using CRISPR-Cas9. This gene-editing system is used to knockout or effectively disable a gene for use by the cell. Then the sample's further survival and proliferation is measured. A negative score implies inhibited success of the cancer sample, suggesting that the gene that was disabled was of use to the cancer. Genes are tested one at a time to remove bias from confounds. In comparison to the above mentioned testing methods, CRISPR-Cas9 is becoming a relatively cheaper option, especially for the non-therapeutic sample testing described above.

Different cancer lineages and subtypes should show distinctly different gene dependency patterns as different areas of the body require different morphological features for a cancer to be successful. This project aims to explore exploiting those differences in order to more easily identify lineage and subtype, or to provide insight into the most probable ones to limit the diagnostic testing needed to be done to confirm the predictions.

Standard of Care and Workflow Integration

As explained above, the current standard for classification does not include this dependency profiling. If integrated into the current system, provided that hospitals have the capacity for this testing available, the new workflow would look like:

- Patient's cancer is replicated across multiple samples, each of which have one of the 500 genes used by this model be a knockout by CRISPR-Cas9.
- Dependency scores are computed for each sample.
- Score profiles are put into the MLP which returns a listing of probable lineages and subtypes.
- The prediction is reviewed by a clinician and standard diagnostic testing will be performed but highly limited as it is just to confirm if the most probable predictions are valid.

3. Technical Description

Input Data

The MLP takes three dataset inputs. Each dataset comes from the DepMap public database. The first is a tabular listing of CRISPR Dependency Scores. Each row represents a singular cancer sample and each column represents one of 500 genes. The 500 genes represent the features of this project. Scores are continuous values ranging from approximately -5 which implies strong dependency by the cancer to 0 or slightly positive which implies non-essential or negative use for the cancer such as remaining tumor suppressors. The 500 genes were chosen to have variety across scores but to be centered around a fairly negative score. Notable cancer genes are included in this model such as KRAS.

This model does contain some missing values, though they tend to be sparse, non-systematic, and of little consequence. For this reason, a median impute is used to fill in the missing data, which is justified by how low the missing scores are (frequently under 5% per gene).

The other two datasets contain the lineage and subtype of the cancers which have a wide range of sources such as lung, breast, lymphoid, etc. Due to some lineages and subtypes being naturally common and others being rare, there are some classes that are too sparse to be used. Classes with less than 5 samples are dropped as it is insufficient data to test on (does not pass the 80-20 split for training and testing data).

Two Classification Tasks

While having been presented in tandem, the two classification tasks are done independently of one another, although they may be related.

- **Lineage prediction:** Predicts the broad tissue of origin. After filtering, approximately 26 different lineages are represented.
- **Subtype prediction:** Predicts a more granular cancer classification (e.g., "High-Grade Serous Ovarian Cancer," "Colon Adenocarcinoma"). After filtering, 61 different subtypes are represented.

Model Architecture

The classifier used in this project is a Multi-Layer Perceptron (MLP) which is a neural network that is well-suited for tabular numerical data like this project's features dataset. The input layer is 500 vectors of gene data with one vector per gene.

This input layer then passes through three hidden layers starting at 512 neurons to match the width of the input layer to 256 neurons to 128 neurons. Starting at 512 neurons allows no initial compression to fully explore connections between gene vectors, giving an opportunity to find meaningful connections before funnelling down with the next layer. Within each hidden layer, four different operations occur. Firstly, a linear transformation is used to apply current weights to the training data. Secondly, batch normalization occurs which rescales values outputted from using the weights which helps to remove noise from the outputs. Thirdly, a ReLU activation is implemented which helps to find non-linear connections between the data. Lastly, a random dropout occurs of 40% of the neurons (standard is 40-50% for MLPs and most neural networks) which forces utilization of multiple pathways rather than a single one that just best describes the training data, preventing over-fitting and learning meaningful connections. Implementing ReLU helps the model overcome the issue of the features being non-linear as genomics typically is; implementing batch normalization and dropout is important as it removes noise. Since some classes have sparse data, the model may over-rely on a single pathway integral to the specific training data leading to over-fitting. Dealing with such behaviors with noise removal overcomes this issue and leads to generalization and external validation.

The output layer uses the final weights from the last hidden layer to then output probabilities of samples being certain lineages/subtypes. From these output values, the model then evaluates the accuracy of these predictions. For each class, CrossEntropy Loss is used to actually convert the raw output into probabilities and then determines the error of the judgements made based on the correctness of the model and the confidence within that decision, meaning correct decisions with low confidence are still penalized. Taking into account both helps push the model to be accurate and well-calibrated for new data encounters.

After the loss is computed, the Adam Optimizer uses the gradient of the loss computed across all weights as well as the gradient squared in order to determine in which direction and how much the weights should be changed from what they previously were, allowing for amplification and inhibition of different weights. This permits for relatively easy maintenance of the high dimensionality of the dataset used. The feedback also decays weights as a feedback, preventing weights from becoming too large and dominant which may cause over-fitting on training data. The Adam optimizer also has an adaptable step size and learning rate compared to other optimization methods, allowing for quicker convergence and learning.

The learning rate scheduler also determines the learning rate on model performance. If the loss is great, the learning rate remains at its initial higher value to explore the data more. If the loss is stagnant at a lower value for a few epochs, then the learning rate gets halved to allow for fine-tuning rather than still being more exploratory.

The main safeguard preventing over-fitting on training data is a patience counter which will stop the learning before the last epoch if the validation scores start to increase which is an indication of over-fitting on training data since the model is not generalizable.

Computational Requirements

The bulk of the data processed is the CRISPR Dependency Scores which track 500 gene dependencies over 1000+ cancer samples. The size of this data is low enough such that the model can be run on a CPU rather than a GPU, although it may take some additional time. The nature of this neural network being run on tabular numeric data also means that it does not strain local memory and RAM much either.

There is a patience of 10 epochs, which means that if no improvement is seen in the validation data testing in 10 epochs, then the model will stop early before the 100 epoch maximum to prevent over-fitting. Most of the runs stopped before the 100 epoch maximum. The model was run 10 times each for lineage and subtype each using a different random seed in order to average metrics across all samples.

Due to the sparse data for some classes in the subtype dataset, SMOTE was used to create augmented data based on the original dataset to see if it would improve performance which creates the largest strain on the computation power of the system.

4. Performance Evaluation

Metrics Explained

Several metrics are used to evaluate the performance of this model for classification of lineage, subtype, and subtype using SMOTE data augmentation.

Because cancer classification involves many classes (not just two), several metrics are used together:

- **Accuracy:** Fraction of samples whose true classification matches the MLP's most probable choice.
- **Top-3 Accuracy:** Fraction of samples whose true classification matches the MLP's top three most probable choices. Clinically matters as it highly limits the pool of potential candidates, permitting streamlined diagnostic testing needing to be done to confirm predictions.
- **F1-Macro:** Computed F1 score (harmonic mean of precision and recall) and averaged across all classes regardless of their frequency in the datasets. Especially shows model weaknesses for rare lineages and subtypes which is a concern due to limited training data.
- **F1-Weighted:** Computed F1 score and averaged across all classes taking into account their frequency making it weighted. Reflects clinical performance since common classes will be encountered more frequently in practice.
- **ROC-AUC (one-vs-rest, weighted):** Measured ability for the MLP to be able to discriminate between the right type vs the rest (all the other types). A value of 0.5 comes from pure chance and 1.0 is perfect discrimination.
- **Matthews Correlation Coefficient (MCC):** Classification metric taking into account TP, TN, FP, and FN and the class imbalances, making it reliable for this highly unbalanced class scenario.

Results (10-run average ± standard deviation)

Metric	Baseline Lineage	MLP Lineage	Baseline Subtype	MLP Subtype	Baseline Subtype + SMOTE	MLP Subtype + SMOTE
Accuracy	0.105±0.001	0.536±0.02	0.056±0.001	0.447±0.02	0.020±0.001	0.470±0.03
Top 3 Accuracy	0.143±0.001	0.734±0.02	0.071±0.001	0.636±0.03	0.035±0.001	0.634±0.02
ROC-AUC	0.500±0.001	0.912±0.01	0.500±0.001	0.907±0.01	0.500±0.001	0.894±0.02
F1-Weighted	0.020±0.001	0.520±0.01	0.006±0.001	0.397±0.01	0.001±0.001	0.427±0.03
F1-Macro	0.007±0.001	0.413±0.02	0.002±0.001	0.257±0.03	0.001±0.001	0.287±0.03
MCC	0.000±0.000	0.510±0.02	0.000±0.000	0.430±0.03	0.000±0.000	0.453±0.03

Baseline: a "most frequent class" dummy classifier that always predicts the most common cancer type.

Interpreting the Results

This MLP highly outperforms the baseline across all metrics for both lineage and subtype. A ROC-AUC score of 0.9 shows that the MLP is able to discriminate between classes very well and reliably while ranking predictions.

For lineage, the MLP gets ~54% accuracy and ~73% top-3 accuracy, against a 10% and 14% baseline. An F1-Macro score of ~0.41 suggests imperfect classification of rare lineages but still a decent performance accounting for 26 classes and the high imbalance.

For subtype, the MLP gets 45–47% accuracy and 64-65% top-3 accuracy depending on whether or not SMOTE is used, against a 2-6% and 4-7% baseline. Interestingly, SMOTE does not improve metrics by much. This suggests that SMOTE may not help rare subtypes much which makes sense as the data augmentation may not be good as it is based on such a small dataset.

Training Dynamics

Training curves (see Appendix) for lineage prediction show that training accuracy climbs to over 90% while validation accuracy plateaus at 60%, with validation loss stabilizing around epoch 40 before the patience counter is maximized and an early stopping occurs to prevent more overfitting. The gap between training and validation performance is a sign of some overfitting taking place which is expected given the modest dataset size relative to the number of parameters. Again this is mitigated via early stopping, halved learning rates, and random neuron dropout during hidden layers.

Most Informative Genes (SHAP Analysis)

SHAP (SHapley Additive exPlanations) is a technique that measures how much each feature, in this project this is each gene, contributed to the model's final predictions. The top genes identified have biologically sound reasonings:

For lineage prediction: GPX4, KRAS, SOD2, SOX10, CCND1, MYC, GRB2.
This list includes well-known oncogenes (KRAS, MYC), lineage-specifying transcription factors (SOX10), and common RTK downstream mutations (GRB2).

For subtype prediction: CCND1, MYC, CDK6, CFLAR, GRB2, KRAS.
This list includes an overlap with the former list but also has other oncogenes such as core cell-cycle regulators (CCND1, CDK6) and apoptosis pathway genes (CFLAR/cFLIP), which vary in importance across cancer subtypes.

These biologically grounded findings increase confidence that the model is finding biologically meaningful connections rather than highly rare chance connections across the whole dataset.

Baseline and Alternative Method Comparison

The only comparison performed in this study is against the most-frequent strategy dummy classifier which the MLP outperforms on all metrics. Compared to previous attempt with Random Forest, it performs significantly better for lineage and subtype (without SMOTE) for most metrics (see Appendix).

Documented Failure Scenarios

- **Rare cancer types:** The MLP was least accurate for rare lineages and subtypes as they had few training examples.
- **Subtype complexity:** As subtypes are specific classifications within a broad lineage, some are closely related which the MLP can have difficulty distinguishing between.

5. Governance Framework

5.1 Site Acceptance Testing

Before deploying or using this tool at any site, the following tests should be done:

- **Environment validation:** Check that all packages are running at most recent versions (read README.md for a comprehensive listing of libraries). Run current script and verify outputs remain at their reported levels.
- **Data format check:** Datasets are not direct from DepMap but are modified versions to provide a way to unify samples across sets. Make sure all new data taken from online or inputted manually follows the format (see /data for an example).
- **Minimum sample check:** If new classes are added, confirm that they each have at least 5 samples of data or else the code will automatically filter them out. Make sure new patient data follows the DepMap ID system and does not overlap with existing patient data.
- **Documentation review:** The clinician should review this field guide, especially the limitations of the MLP and the scope to which it should be used.

5.2 Monitoring Strategy

Since this is an investigational research tool, formal ongoing monitoring should follow research best practices:

Monitored Measure	Alert Threshold	Frequency
Overall test accuracy (when ground truth is available)	Drop >10 percentage points	Per experiment batch
F1-Macro	Drop below 0.25	Per experiment batch
MCC	Drop below 0.30	Per experiment batch
Distribution of input gene scores	Mean per gene deviates by >2 standard deviations from training mean	Each new dataset
Class distribution	Any new class not seen during training	Before each run (run code until EDA, can modify in main)

Monitoring procedure: Compare metrics if adding new datasets or batches to the referenced trends above to catch drifts and re-evaluate model use for this new data.

5.3 Escalation Procedures and Contacts

Issue	Action	Contact
Metrics drops below above thresholds	Suspend use and investigate data files	Project team
Unexpected input data format errors	Review data files, especially format	Member handling data entry
Model crashes or produces NaN outputs	Log the error and report bug to developers	Project developers
Potential misuse or use outside intended scope	Alert PI/department head immediately	PI or Department Head

All issues should be filed with the date, dataset description, error type and message with supporting documentation, and resolution if reached. Logs should be retained to prevent future occurrences and to provide better oversight in the future listing common errors.

5.4 Human Oversight Requirements

As stated, this MLP must not be used as a standalone diagnostic instrument in its current state and for the foreseeable future. Human oversight is required at all times for the following:

- **Mandatory review:** Model outputs should be reviewed by clinician with sufficient expertise to verify probable validity before future decisions are made.
- **Corroborating evidence:** Predictions should be followed by subsequent testing with histopathology, molecular sequencing, and IHC data (although testing can be lessened as it is to confirm predictions). Conflicts between prediction and diagnostic testing results require additional investigation. The resolution will most likely be relying on traditional methods to determine the cancer lineage and subtype for the specific case.
- **Investigational use only:** This MLP is not FDA-cleared or CE-marked. It may provide insight but should not make treatment decisions and should not be used outside approved clinical sites and users.
- **Uncertainty acknowledgment:** Always keep in mind error bar uncertainties and model confidence in predictions when using.

6. Limitations and Risks

Known Constraints

- **Cell line data only:** This model was trained on cancer cell lines from DepMap and it is hard to find where every piece of data originated from. If the data came from cells replicated for years in a research lab before being tested on in this way, it may differ from samples taken directly from a patient. This may impact generalizability.
- **500-gene panel:** The model uses only 500 genes due to timing, convergence, and accessibility. Although it was chosen to be a good representation for cancer studies, the full human genome contains ~20,000 genes so the 500 may not cover all relevant aspects.
- **Modest dataset size:** There is limited data available for rare lineages and subtypes which may impact performance for these classes.
- **CRISPR screening issues:** Although CRISPR-Cas9 is becoming increasingly cheaper and easier to perform, to test 500 genes and future proliferation will take time, limiting immediate utility for clinical use.

Populations with Potentially Degraded Performance

- **Rare cancer types:** Classes with fewer training examples (even after SMOTE as it did not improve metrics much) will have less reliable predictions. Extra caution should be taken for lineages or subtypes with a small amount of data.
- **Pediatric cancers:** As of now, DepMap data is predominantly derived from adult cancers. This makes performance for pediatric cancers untested and uncertain. However, DepMap is starting to include more pediatric cancer data, so this can be updated in the future when sufficient data is added.
- **Cancers not represented in DepMap:** A patient with a cancer whose lineage and subtype is not one in the dataset will be misclassified.
- **Samples with high missing data rates:** Genes with a large amount of missing data may be biased with median imputed values.

Misuse Potential

- **Overreliance in clinical settings:** The most critical risk is a clinician treating the MLP's predictions as a confirmed diagnosis rather than one data point or suggestion to follow up on with standard diagnostic testing. This could lead to incorrect classification which may cause inappropriate treatment.
- **Inappropriate population generalization:** Applying the model to patient populations (e.g., fresh biopsies, certain social backgrounds) systematically underrepresented in DepMap could yield biased results.
- **Gaming or adversarial input:** In a research context, mislabeled or corrupted CRISPR data used will produce confident but incorrect outputs and this model does not have a data quality check besides EDA and simple counter filters.

Contraindications for Use

Do not use this tool:
- As the only determination for future patient decisions.
- On samples that have scores for less than 80% of the genes listed.
- On cancer whose suspected lineages and subtypes are not represented in the training data.
- Without access to diagnostic testing (histopathology, molecular profiling) for result validation.
- Outside of an institutional research or clinical setting, especially without an IRB.

7. Appendix

A. Summary of Output Visualizations

File	Description
outputs/gene_correlation_heatmap.png	Heatmap of correlations among a random subset of 50 genes. Tend to be low on average, indicating low data leakage though higher correlations exist between genes of the same family.
outputs/crispr_histogram.png	Distribution of all CRISPR dependency scores. Centered on negative scores with a slight leftwards skew, meaning most genes are important for cancer.
outputs/lineage_histogram.png	Frequency distribution of cancer lineages showing class imbalance.
outputs/subtype_histogram.png	Frequency distribution of cancer subtypes showing more severe class imbalance, leading to the decision to use SMOTE.
outputs/crispr_lineage_dependency.png	Heatmap of mean CRISPR dependency scores per gene, grouped by lineage. Different lineages have different dependencies, validating the mindset behind the model.
outputs/crispr_subtypes_dependency.png	Heatmap of mean CRISPR dependency scores per gene, grouped by subtype. Different subtypes have different dependencies, validating the mindset behind the model.
outputs/results_with_errorbars.png	Bar chart comparing all six metrics across all three experiments (Lineage, Subtype, Subtype+SMOTE) for both the MLP and baseline model, with standard deviation error bars from 10 runs per model.
outputs/shap_lineage_top20.png	Top 20 most influential genes (features) for lineage prediction via SHAP.
outputs/shap_subtype_top20.png	Top 20 most influential genes (features) for subtype prediction via SHAP.
outputs/training_curves_lineage.png	Training and validation loss and accuracy curves for lineage prediction. Shows model convergence, overfitting, and early stopping.
outputs/training_curves_subtype.png	Training and validation loss and accuracy curves for subtype prediction. Shows model convergence, overfitting, and early stopping.
outputs/comparison_rf_mlp.png	Comparison of RF and MLP metrics. Shows significant improvement with MLP.

B. Glossary for Non-ML Readers

Term	Plain-language definition
CRISPR dependency score	A number measuring how much a cancer cell needs a specific gene to survive. More negative = more dependent.
Neural network / MLP	A mathematical system loosely inspired by the brain, composed of many simple mathematical operations ("neurons") stacked in layers, that learns patterns from examples. See Model Architecture for more information.
Training / testing split	The practice of dividing data into one set to teach the model (training) and a separate set it has never seen, used to evaluate real-world performance (testing).
Overfitting	When a model learns training examples too precisely and performs worse on new data — like memorizing answers without understanding the concepts. Limits generalizability/external validation.
Dropout	A training technique that randomly "turns off" neurons to prevent overfitting.
SMOTE	A technique that creates augmented examples of underrepresented classes in an attempt to make up for the low amount of data present.
SHAP	A method for explaining which input features (genes) most influenced a model's prediction for a given sample.

C. Hyperparameter Summary

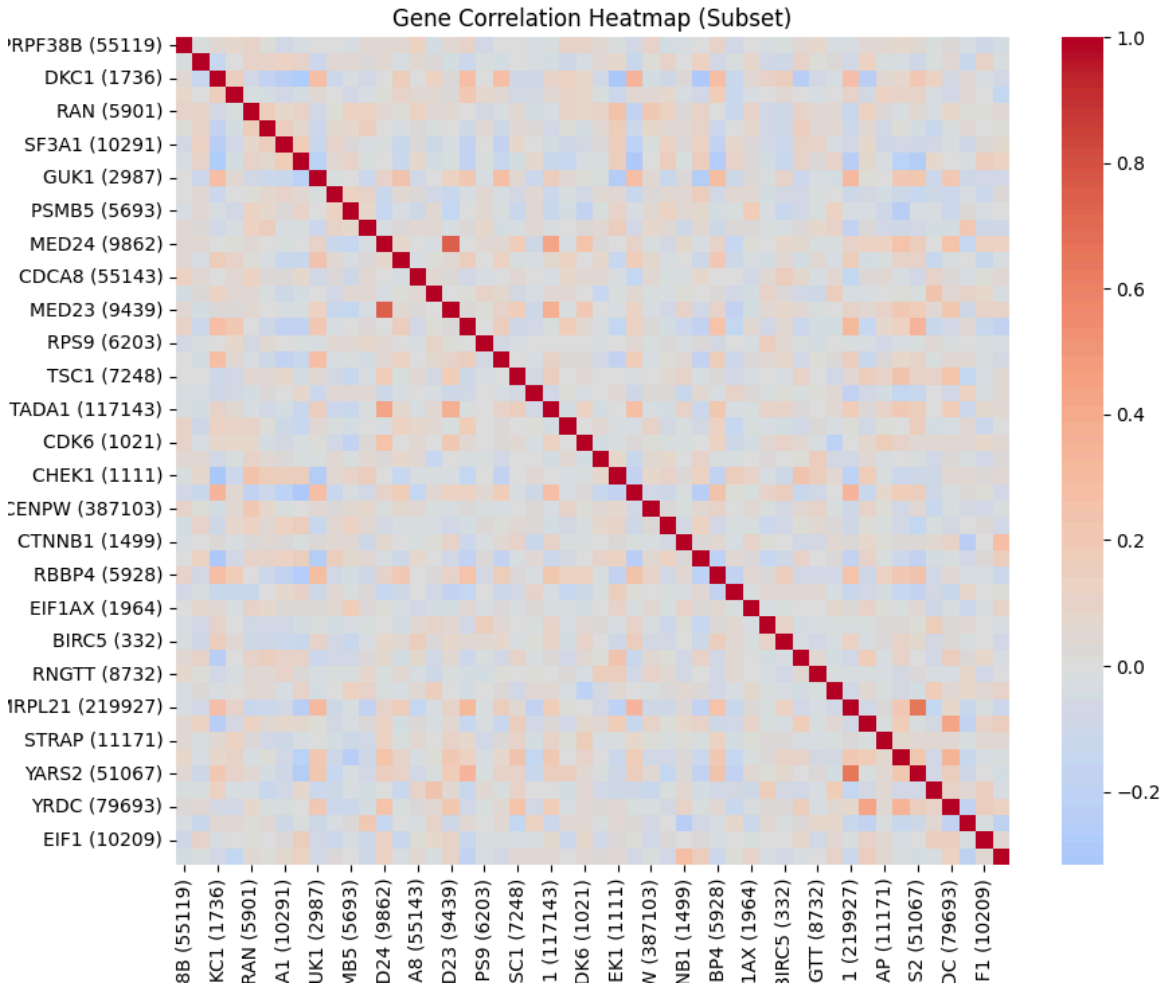
Parameter	Value	Rationale
Hidden layers	[512, 256, 128]	Progressively reduces dimensionality; sufficient capacity for 500 gene inputs
Dropout rate	0.40	High regularization needed given small dataset
Learning rate	0.001	Standard Adam default; scheduler handles decay
Weight decay	0.0001	L2 regularization to further reduce overfitting
Batch size	32	Standard for tabular biological data
Max epochs	100	With early stopping, rarely reached
Early stopping patience	10 epochs	Prevents premature stopping on noisy val loss
LR scheduler patience	5 epochs	Halves LR when val loss plateaus
SMOTE k-neighbors	1	Conservative; small k appropriate for sparse subtype classes
Minimum class size	5 samples	Ensures each class has enough examples for stratified splitting
Experiment repetitions	10 runs	Produces stable mean and standard deviation estimates

D. Recommended Future Work

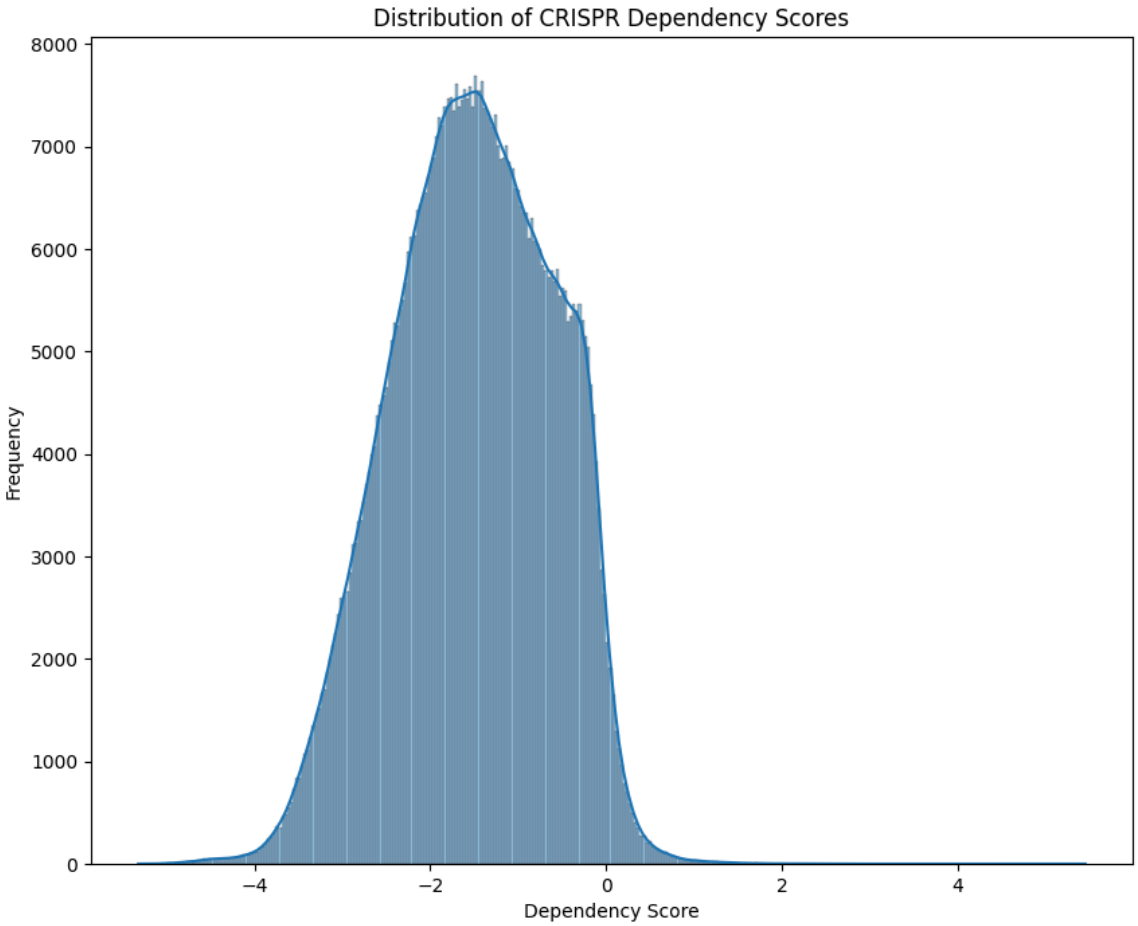
- **Comparison to other classifiers:** Evaluate Random Forest, Support Vector Machines, and gradient-boosted trees (XGBoost) on the same data to contextualize MLP performance.
- **External validation:** Test on a held-out dataset from a different CRISPR screening platform or laboratory to assess true generalizability.
- **Per-class performance breakdown:** Report accuracy and F1 scores for each individual lineage and subtype to identify where the model is strongest and weakest.
- **Patient-derived sample testing:** Validate whether models trained on cell lines can predict cancer type in patient-derived organoids or freshly isolated tumor cells.
- **Demographic subgroup analysis:** Once patient-derived data is available, assess whether performance differs by patient age, sex, or ancestry.
- **Uncertainty quantification:** Implement prediction confidence intervals (e.g., Monte Carlo Dropout or conformal prediction) to allow users to identify low-confidence predictions in real time.

E. Visualizations

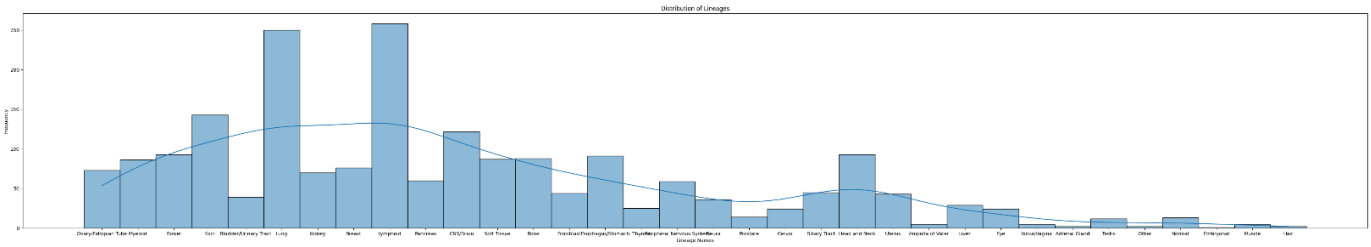
outputs/gene_correlation_heatmap.png



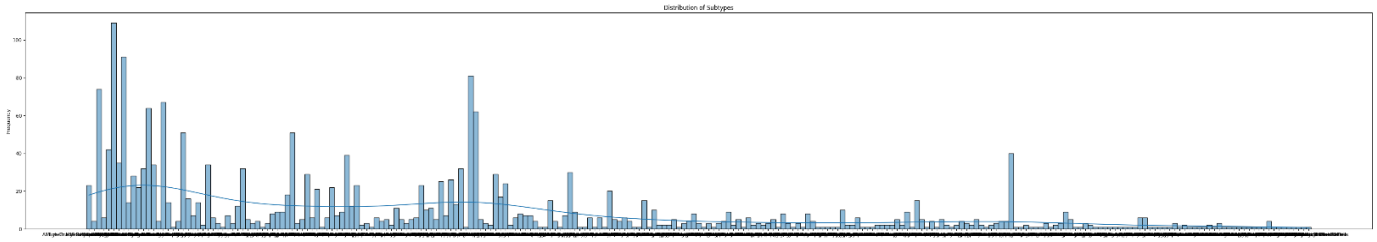
outputs/crispr_histogram.png



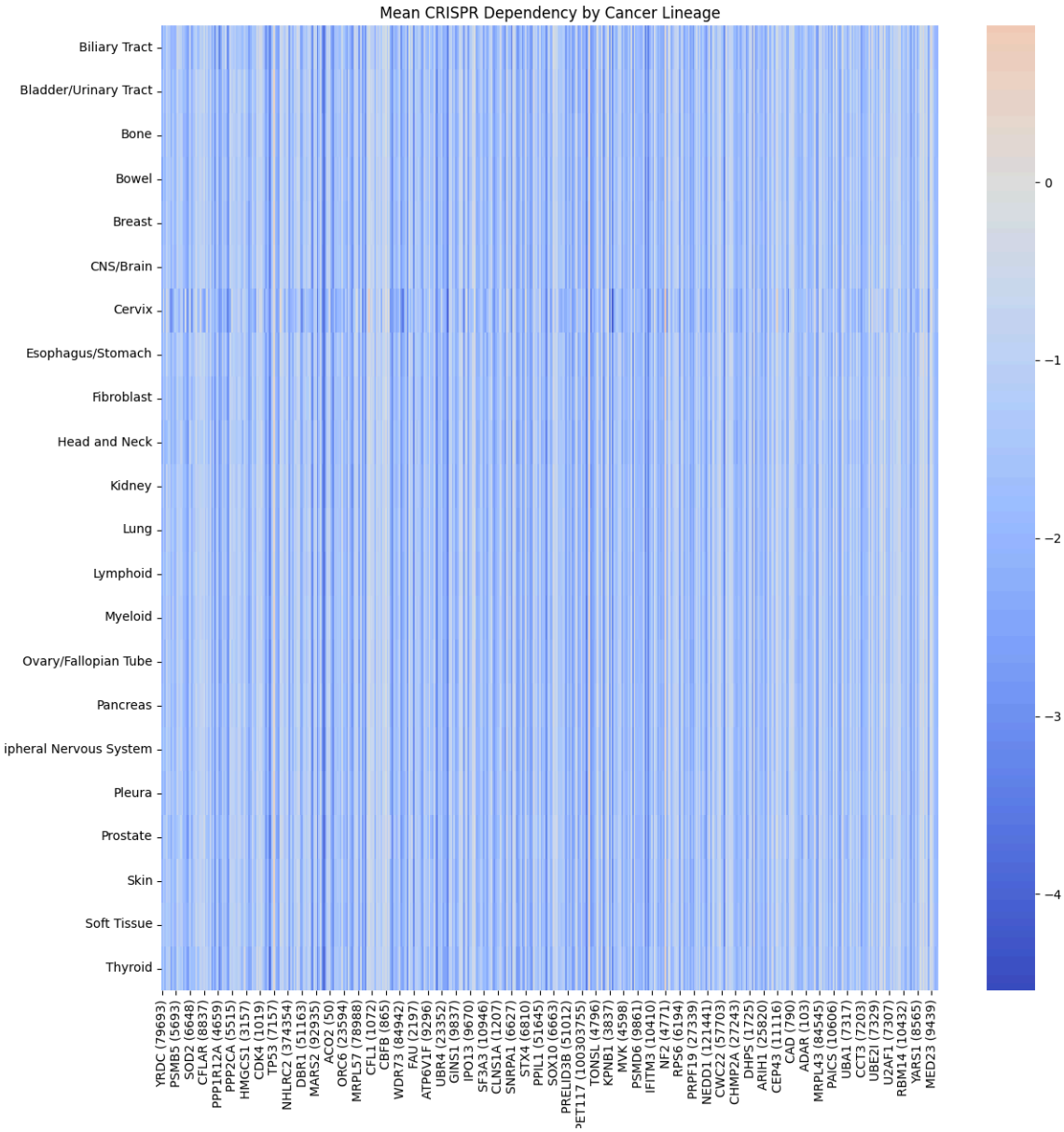
outputs/lineage_histogram.png



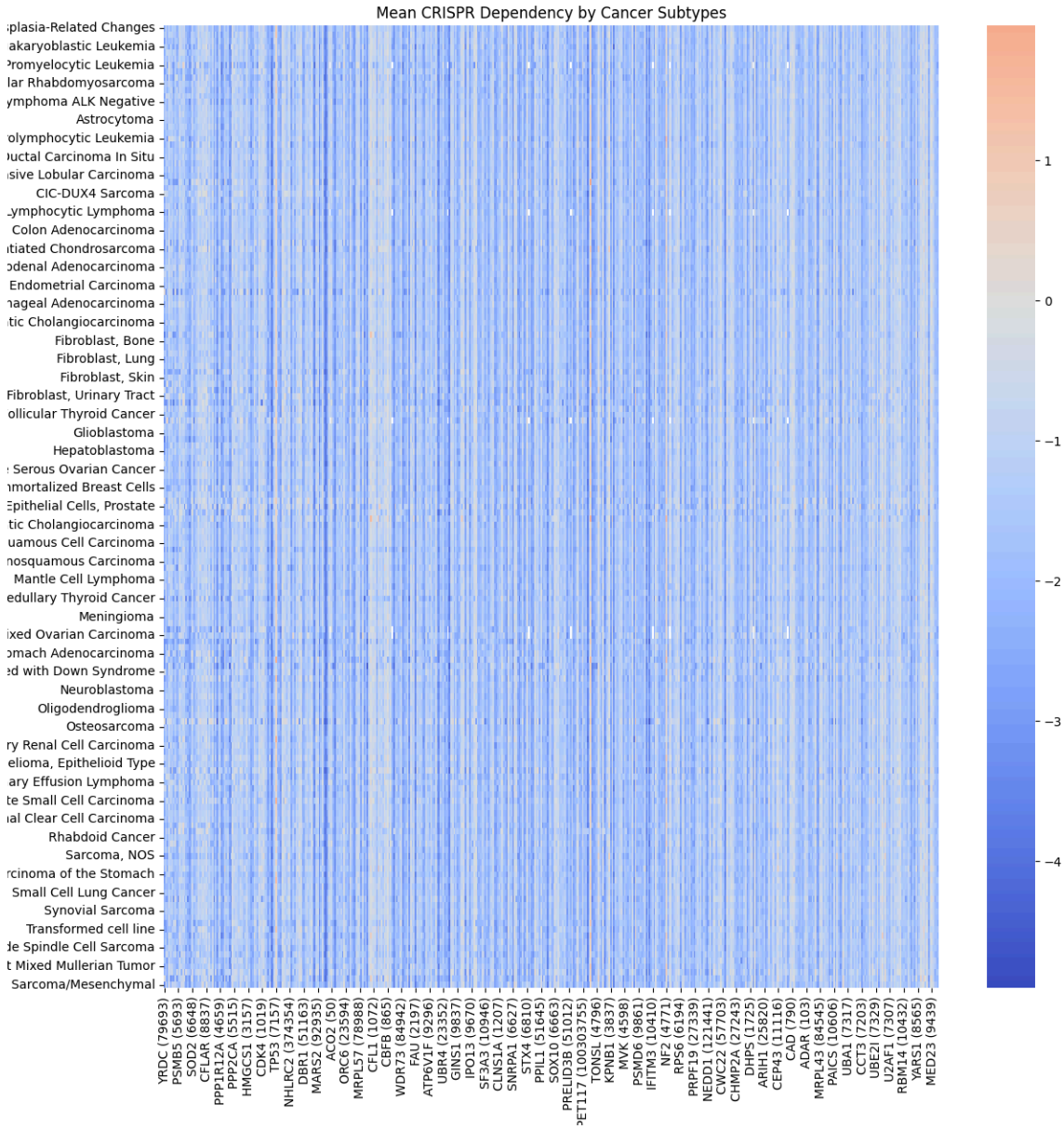
outputs/subtype_histogram.png



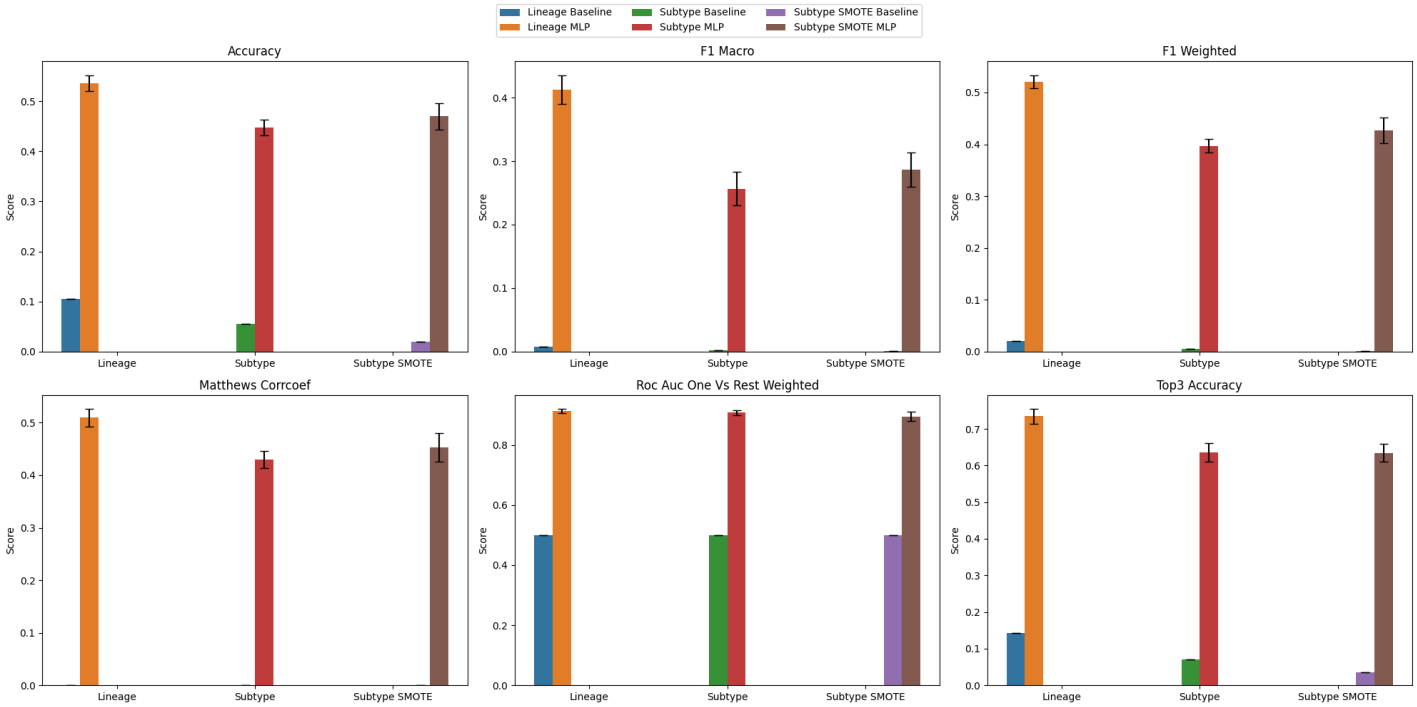
outputs/crispr_lineage_dependency.png



outputs/crispr_subtypes_dependency.png

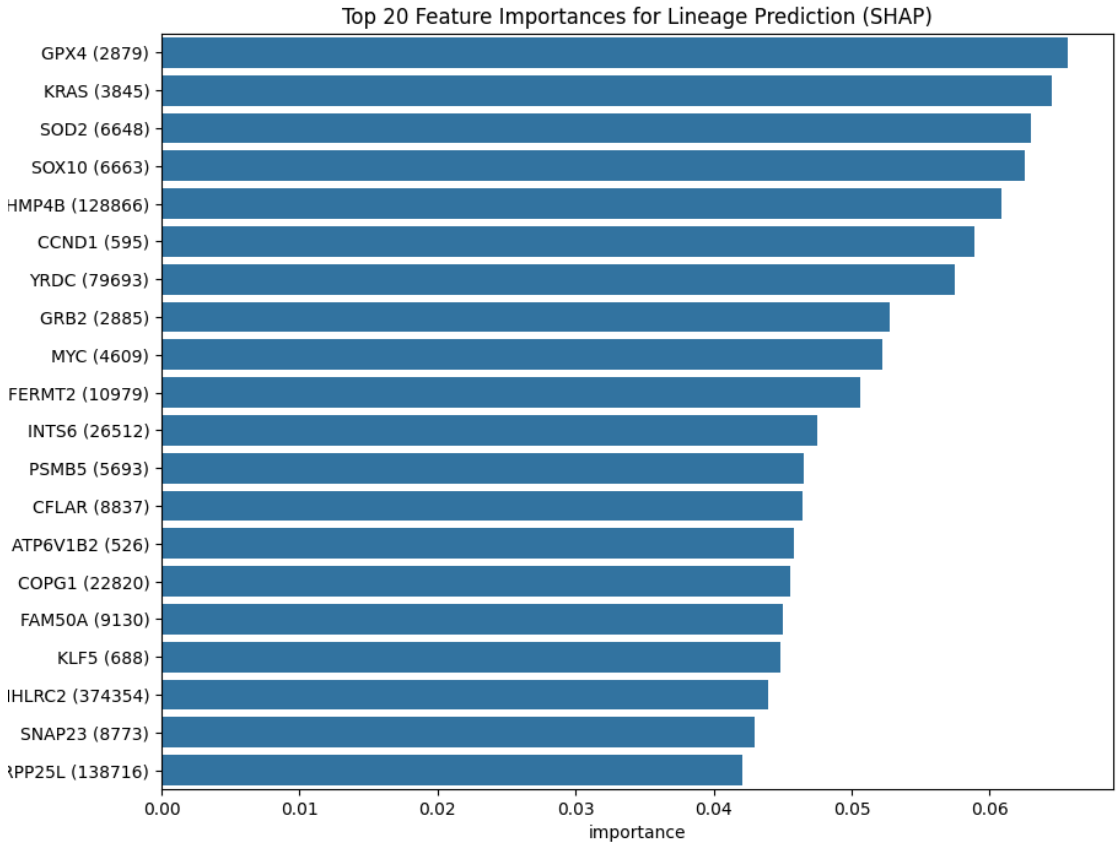


outputs/results_with_errorbars.png

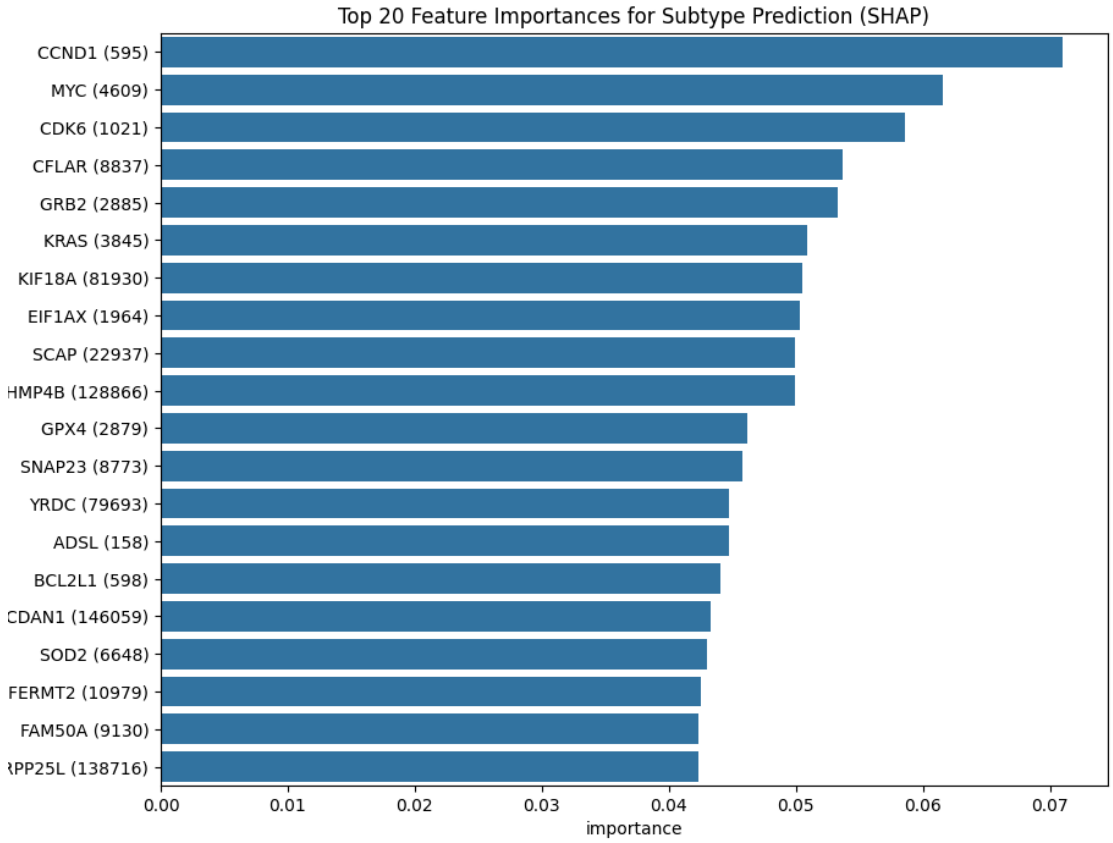


Metric	Baseline Lineage	MLP Lineage	Baseline Subtype	MLP Subtype	Baseline Subtype + SMOTE	MLP Subtype + SMOTE
Accuracy	0.105±0.001	0.536±0.02	0.056±0.001	0.447±0.02	0.020±0.001	0.470±0.03
Top 3 Accuracy	0.143±0.001	0.734±0.02	0.071±0.001	0.636±0.03	0.035±0.001	0.634±0.02
ROC-AUC	0.500±0.001	0.912±0.01	0.500±0.001	0.907±0.01	0.500±0.001	0.894±0.02
F1-Weight ed	0.020±0.001	0.520±0.01	0.006±0.001	0.397±0.01	0.001±0.001	0.427±0.03
F1-Macro	0.007±0.001	0.413±0.02	0.002±0.001	0.257±0.03	0.001±0.001	0.287±0.03
MCC	0.000±0.000	0.510±0.02	0.000±0.000	0.430±0.03	0.000±0.000	0.453±0.03

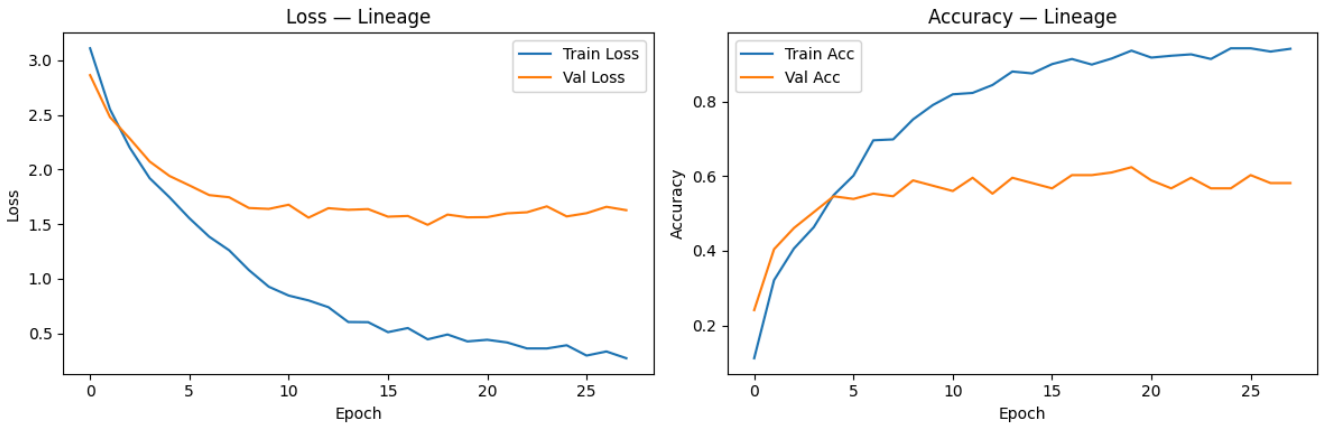
outputs/shap_lineage_top20.png



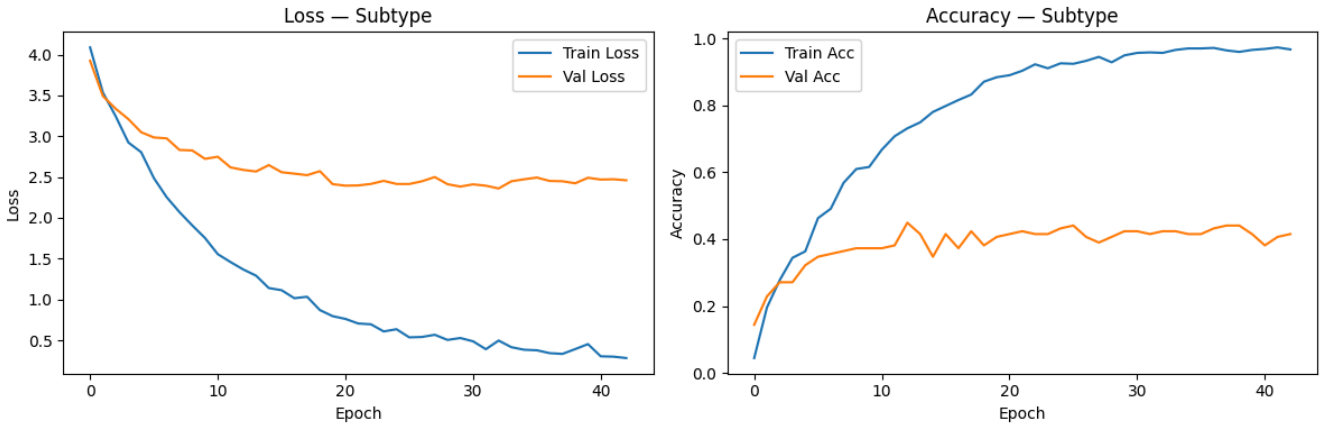
outputs/shap_subtype_top20.png



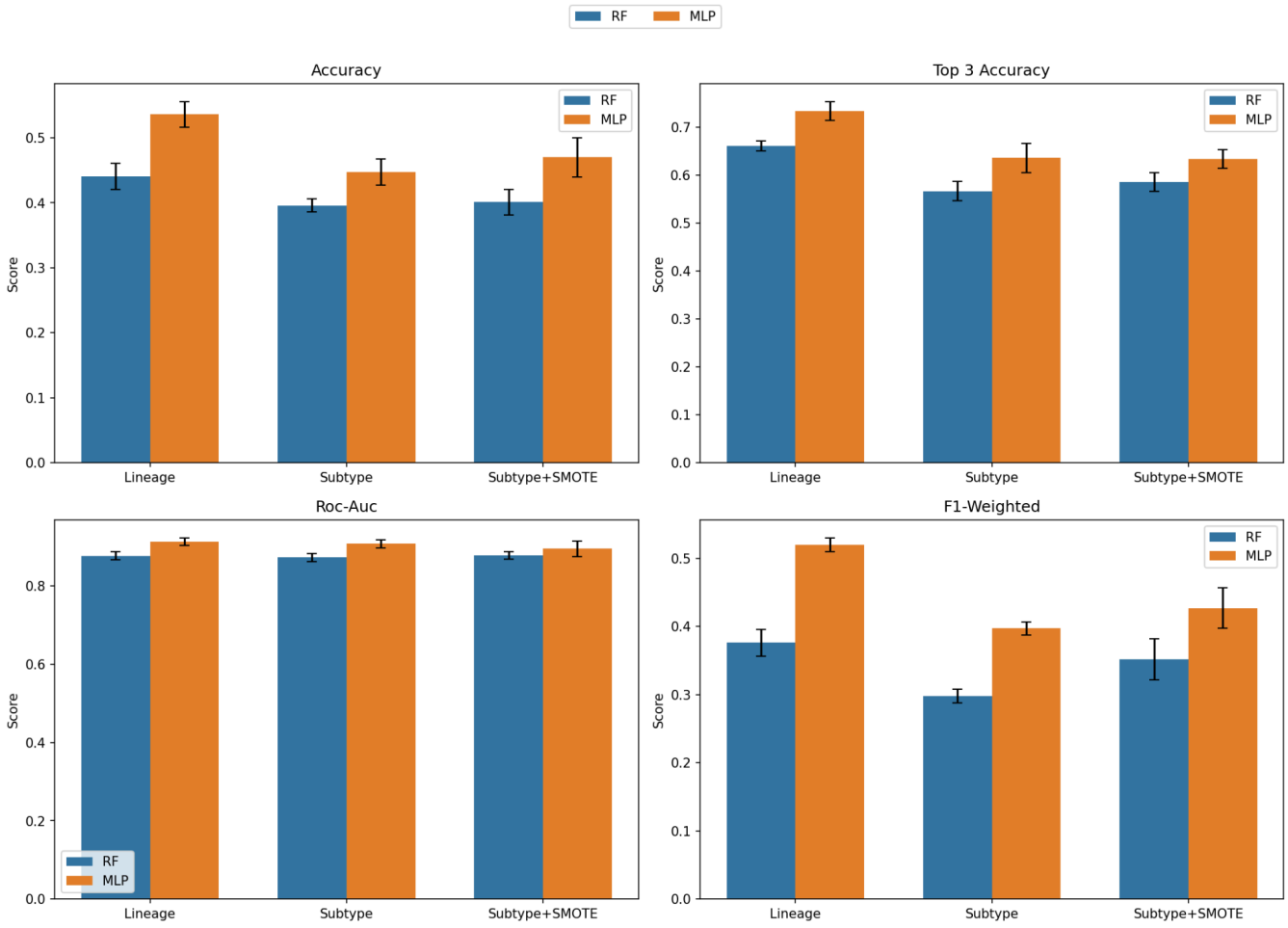
outputs/training_curves_lineage.png



outputs/training_curves_subtype.png



outputs/comparison_rf_mlp.png


RF
MLP
Accuracy
Top 3 Accuracy
Roc-Auc
F1-Weighted
Score
Lineage
Subtype
Subtype+SMOTE